School of Future Tech

Case Study Report

on

The Transport Billing & Invoice Management System

by

YATHARTH DHARMESH MISHRA

150096725055

INDEX

1. Introduction to the Case Study
2. Problem Statement / Case Background (Abstract)
3. Problem Statement / Case Study Design
4. Methods & Algorithms Technology Applied
5. Implementation Details and Snapshots
6. Results and Conclusion
7. References

---

1. Introduction to the Case Study

The Transport Billing & Invoice Management System is a console-based C++ application developed to automate the billing process in transport businesses. Transport companies often generate invoices manually, which may lead to calculation errors, mismanagement of payment records, and loss of important data.

This system helps in accurate bill calculation, invoice generation, and payment tracking using object-oriented programming concepts and file handling in C++.

---

2. Problem Statement / Case Background (Abstract)

In small and medium-scale transport businesses, invoice generation is often handled manually. This results in:

- Calculation mistakes
- Incorrect billing amounts
- Poor payment tracking
- Data loss
- Time-consuming processes

The objective of this project is to develop a Transport Billing & Invoice Management System that:

- Calculates transport charges automatically

- Generates invoices accurately
- Tracks payment status
- Stores records permanently using file handling

---

3. Problem Statement / Case Study Design

The system is designed using Object-Oriented Programming (OOP) concepts.

System Components:

- User Input Module
- Billing Calculation Module
- Invoice Generation Module
- Payment Tracking Module
- File Storage Module

Class Used:

`Transport` class

Data Members:

- Customer Name
- Distance (km)
- Rate per km
- Load Weight
- Total Bill
- Payment Status

Member Functions:

- inputDetails()
- calculateBill()
- generateInvoice()
- makePayment()
- saveToFile()

---

4. Methods & Algorithms Technology Applied

Technologies Used:

- C++ Programming Language
- Object-Oriented Programming
- File Handling (`ofstream`)
- Menu-Driven System

Algorithm for Billing Calculation:

Step 1: Input customer details
Step 2: Calculate transport charge
Step 3: Add weight-based charge
Step 4: Compute total bill
Step 5: Set payment status as "Pending"
Step 6: Generate invoice
Step 7: Save details into file

Formula Used:

Total Bill = (Distance × Rate per km) + (Load Weight × 50)

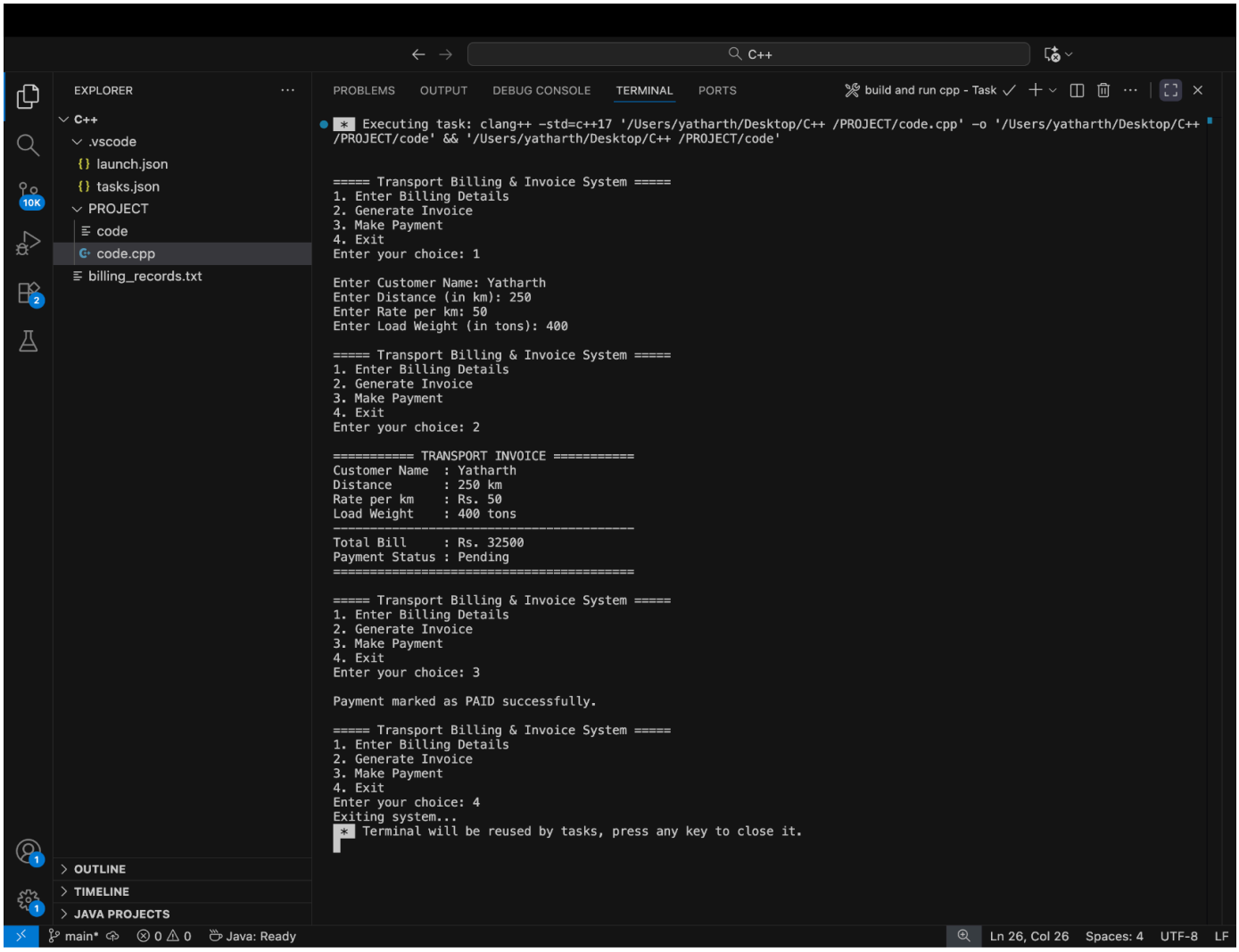
---

5. Implementation Details and Snapshots

Implementation Details:

- The system uses a class to manage billing data.
- File handling is used to store billing records permanently.
- A menu-driven interface allows users to choose operations.
- Payment status can be updated from Pending to Paid.

Sample Output Snapshot:



---

6. Results and Conclusion

The Transport Billing & Invoice Management System successfully automates the billing process. It reduces manual calculation errors and improves record maintenance through file storage.

Advantages:

- Accurate billing calculation
- Faster invoice generation
- Proper payment tracking
- Permanent data storage

- Easy to use menu-driven interface

Conclusion:

The system meets all functional requirements such as billing calculation, invoice generation, and payment tracking. It demonstrates practical implementation of Object-Oriented Programming and file handling concepts in C++.

---

7. References

1. Bjarne Stroustrup – The C++ Programming Language
2. C++ Documentation – www.cplusplus.com
3. Object-Oriented Programming Concepts – Academic Notes